Assignment 5: Data Visualization

Meg O'Brien

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

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Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1

library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```
library(ggplot2)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
library(lubridate)
library(viridis)
```

```
## Loading required package: viridisLite
```

```
library(RColorBrewer)
library(colormap)

getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```
NTL_LTER <- read.csv(
  file = here
('Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv')
, stringsAsFactors = TRUE)

NEON_NIWO <- read.csv(
  file = here('Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv'), stringsAsFactors = TRUE)

#2

class(NTL_LTER$sampledate)
```

```
## [1] "factor"
```

```
class(NEON_NIWO$collectDate)
```

```
## [1] "factor"
```

```
#Changing factor classes to dates

NTL_LTER$sampledate <- as.Date(NTL_LTER$sampledate, format = "%Y-%m-%d")
NEON_NIWO$collectDate <- as.Date(NEON_NIWO$collectDate, format = "%Y-%m-%d")
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels

- Axis ticks/gridlines
- Legend

```r
#3

mytheme <- theme_cowplot()+
  theme(
    plot.background = element_rect(fill = "mistyrose", color = NA),
    axis.text = element_text(color = "black"),
    legend.position = "top")

theme_set(mytheme)
```

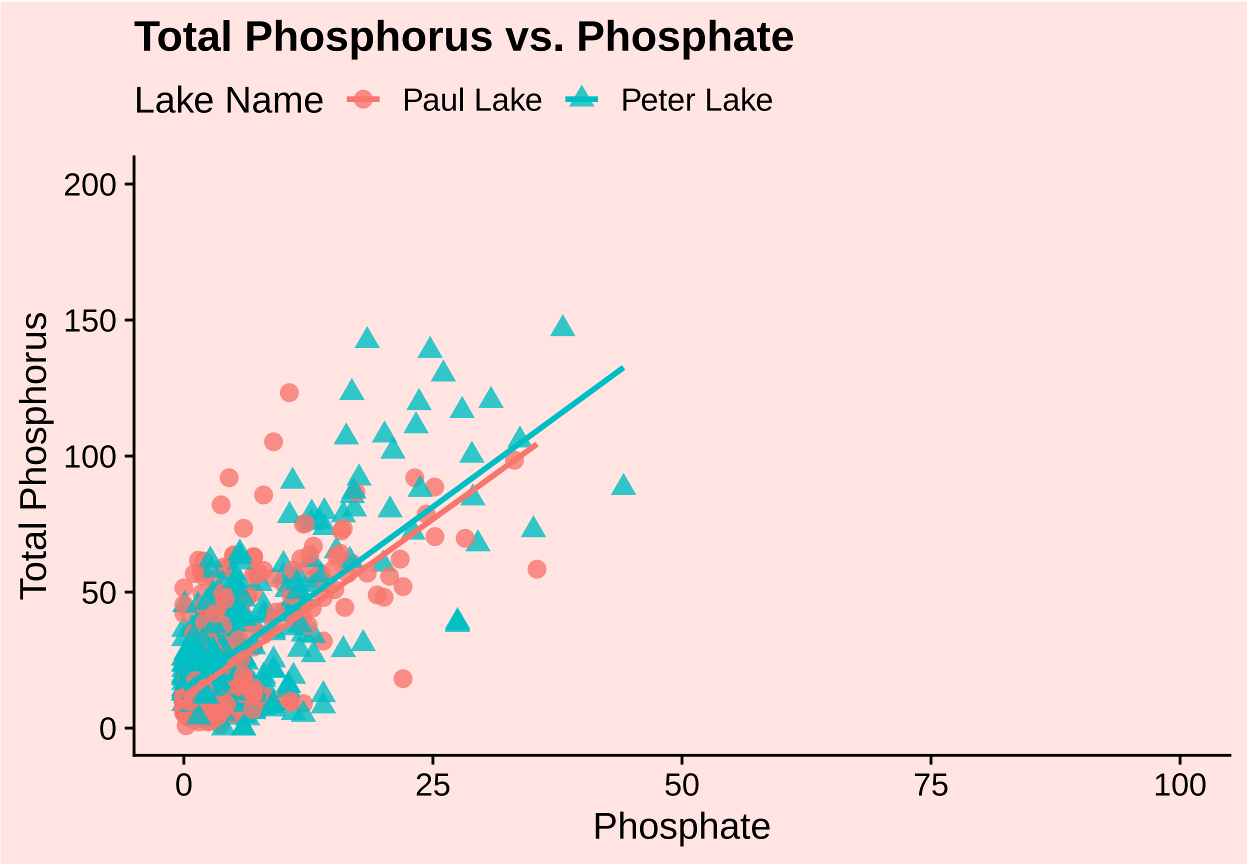
Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

```r
#4

P_vs_PO4 <- ggplot(NTL_LTER, aes(x = po4, y = tp_ug, color = lakename,
                                 shape = lakename))+
  geom_point(size = 3, alpha = 0.8) +
  geom_smooth(method=lm,se=FALSE) +
  labs(
    title = "Total Phosphorus vs. Phosphate",
    x = "Phosphate",
    y = "Total Phosphorus",
    color = "Lake Name",
    shape = "Lake Name"
  ) +
  xlim(0, 100) +
  ylim(0, 200)
print(P_vs_PO4)
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 21948 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21948 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



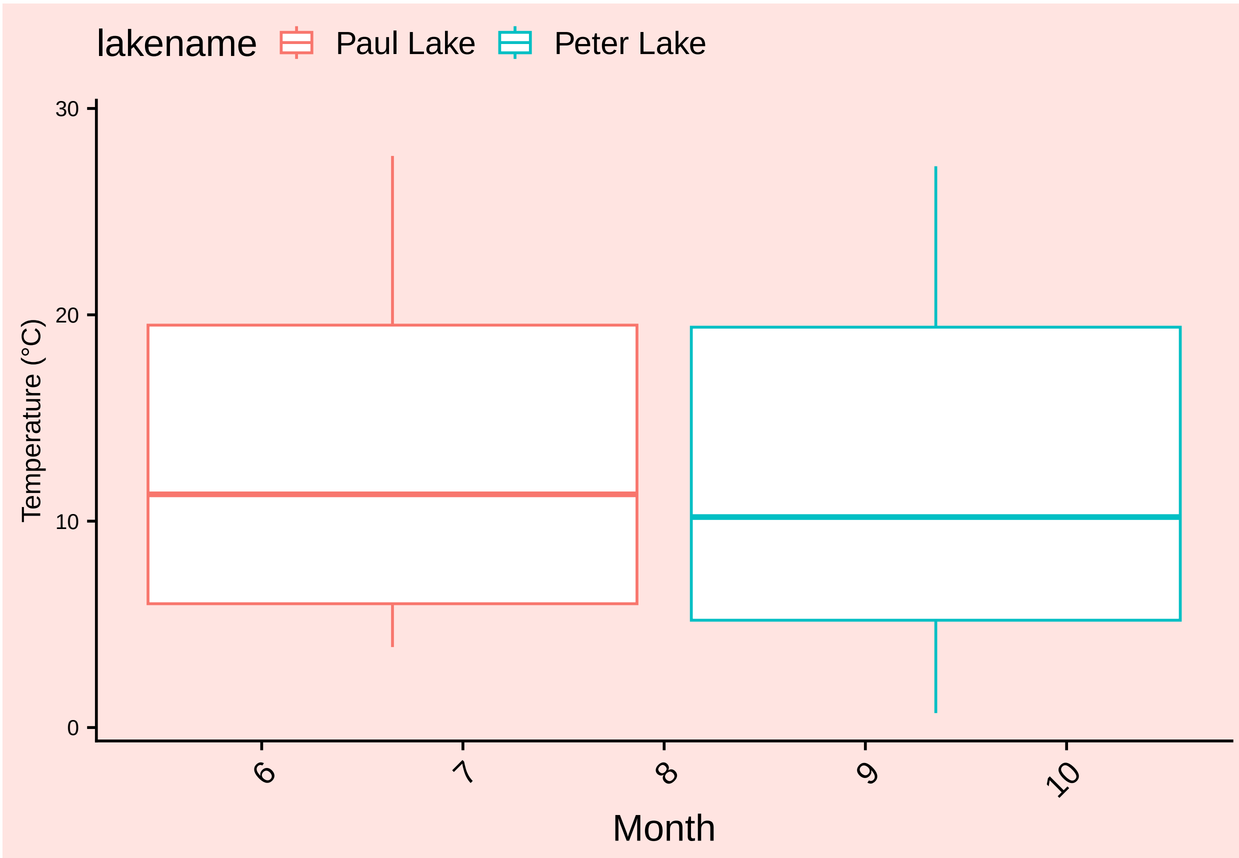
5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and

(c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

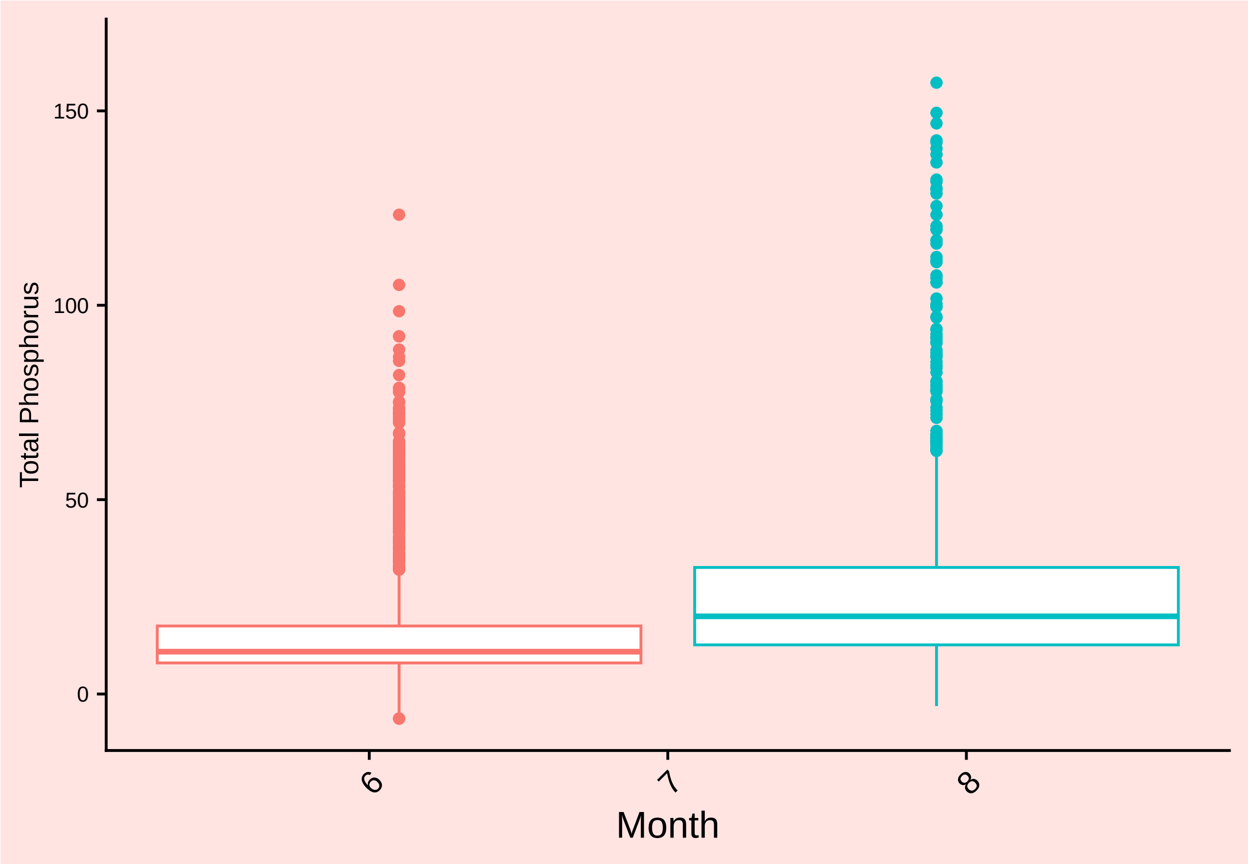
```
#5

#a) Temperature Plot
Temp_plot <- ggplot(NTL_LTER, aes(x = month, y = temperature_C,
                                  color = lakename))+
  geom_boxplot()+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.1))) +
  labs(y = "Temperature (°C)", x = "Month") +
  mytheme +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "top",
    axis.title.y = element_text(size = 10),
    axis.text.y = element_text(size = 8)
  )
print(Temp_plot)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



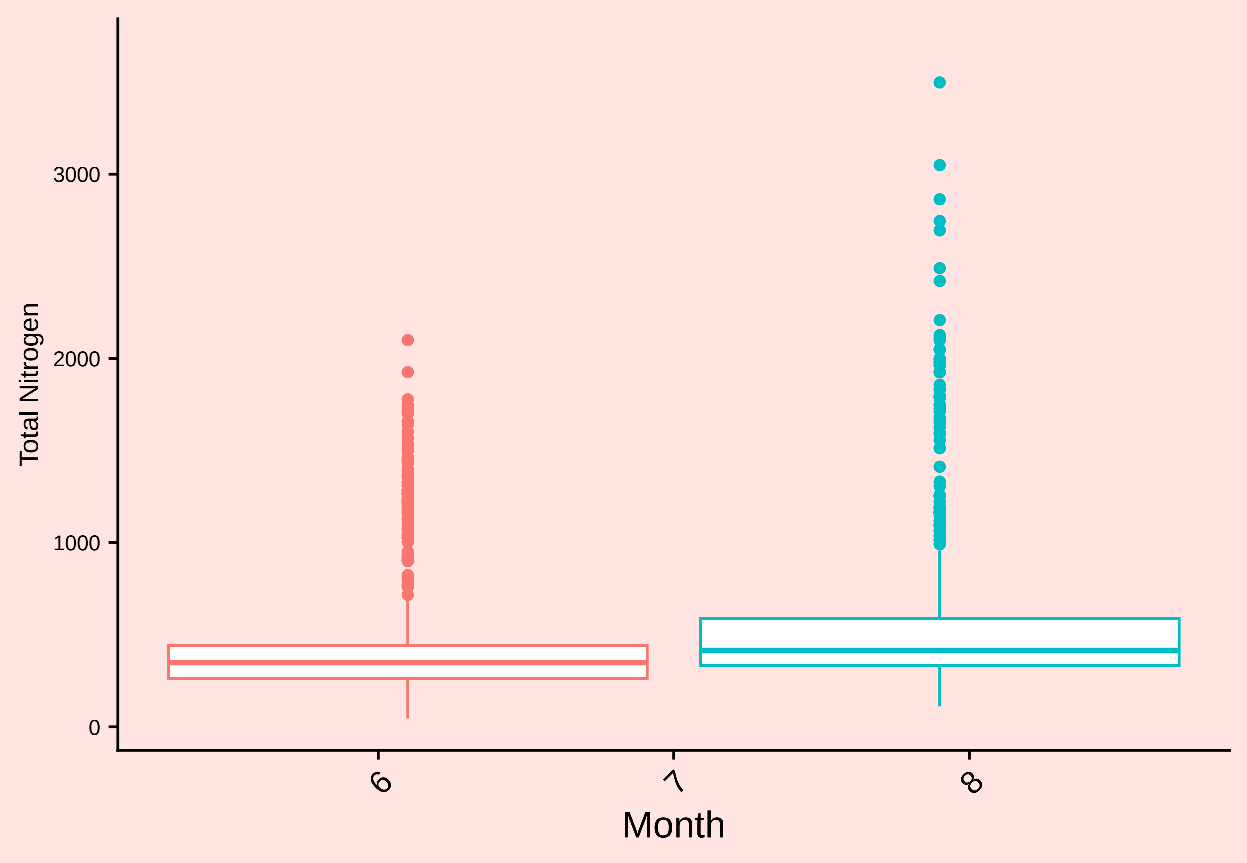

```
#b) TP Plot

TP_plot <- ggplot(NTL_LTER, aes(x = month, y = tp_ug, color = lakename))+
  geom_boxplot()+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.1))) +
  labs(y = "Total Phosphorus", x = "Month") +
  mytheme +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    axis.title.y = element_text(size = 10),
    axis.text.y = element_text(size = 8)
  )
print(TP_plot)
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```r
#c) TN Plot

TN_plot <- ggplot(NTL_LTER, aes(x = month, y = tn_ug, color = lakename))+
  geom_boxplot()+
  scale_y_continuous(expand = expansion(mult = c(0.05, 0.1))) +
  labs(y = "Total Nitrogen", x = "Month") +
  mytheme +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    axis.title.y = element_text(size = 10),
    axis.text.y = element_text(size = 8)
  )
print(TN_plot)
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#Combine Plots

legend <- get_legend(Temp_plot)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```

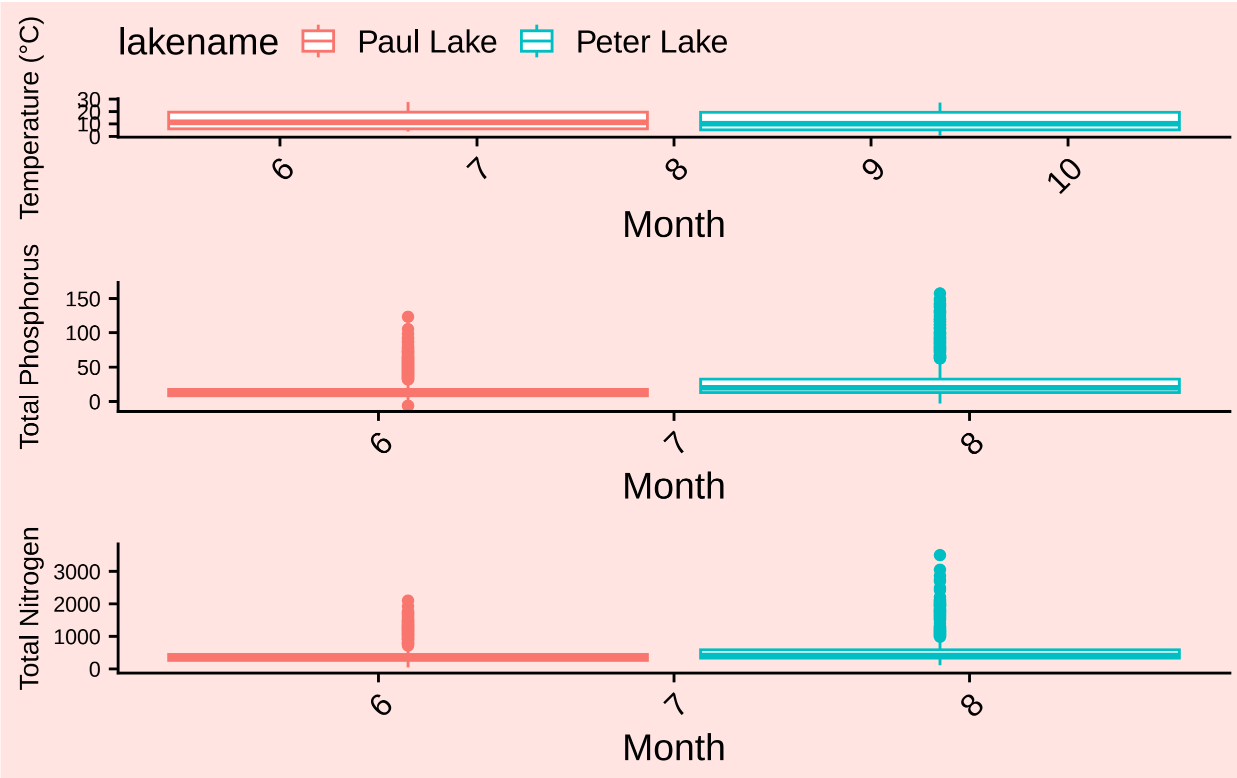
```
combined <- plot_grid(
  Temp_plot, TP_plot, TN_plot,
  ncol = 1,
  align = "v"
)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 20729 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 21583 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```r
final_combined <- plot_grid(combined, legend, ncol = 1, rel_heights = c(1, 0.1))
print(final_combined)
```



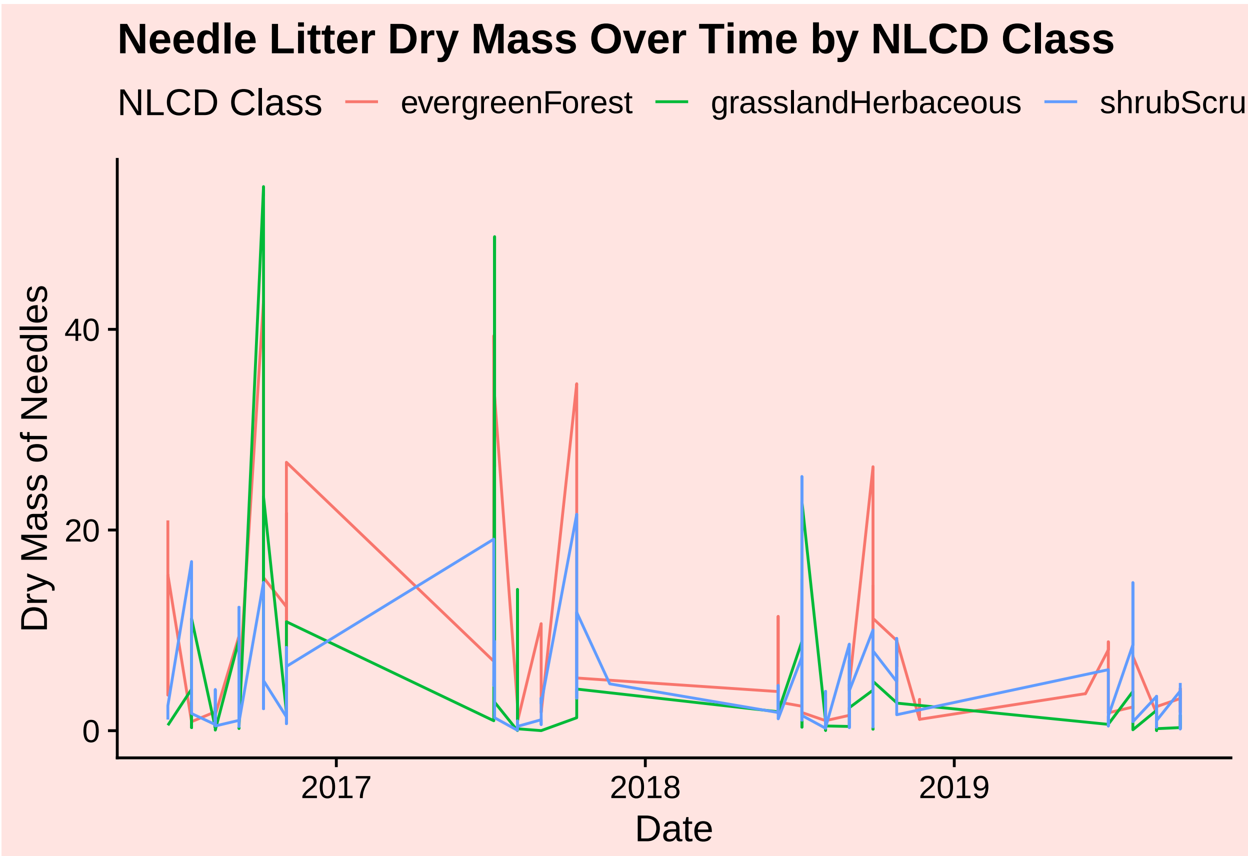
Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: Based on the observed boxplots of Peter and Paul Lake for temperature, total phosphorus and total nitrogen over the sampling months, it is clear that Paul and Peter Lakes were sampled within different months of the year. Paul Lake was sampled during the firt half of the year and Peter Lake was sampled during the second half of the year. Also, it is evident that Peter and Paul Lake have comprable temperatures, with Peter Lake showing slightly greater temperature variation. Peter Lake also shows higher total nitrogen and total phosphorus levels than Paul Lake.

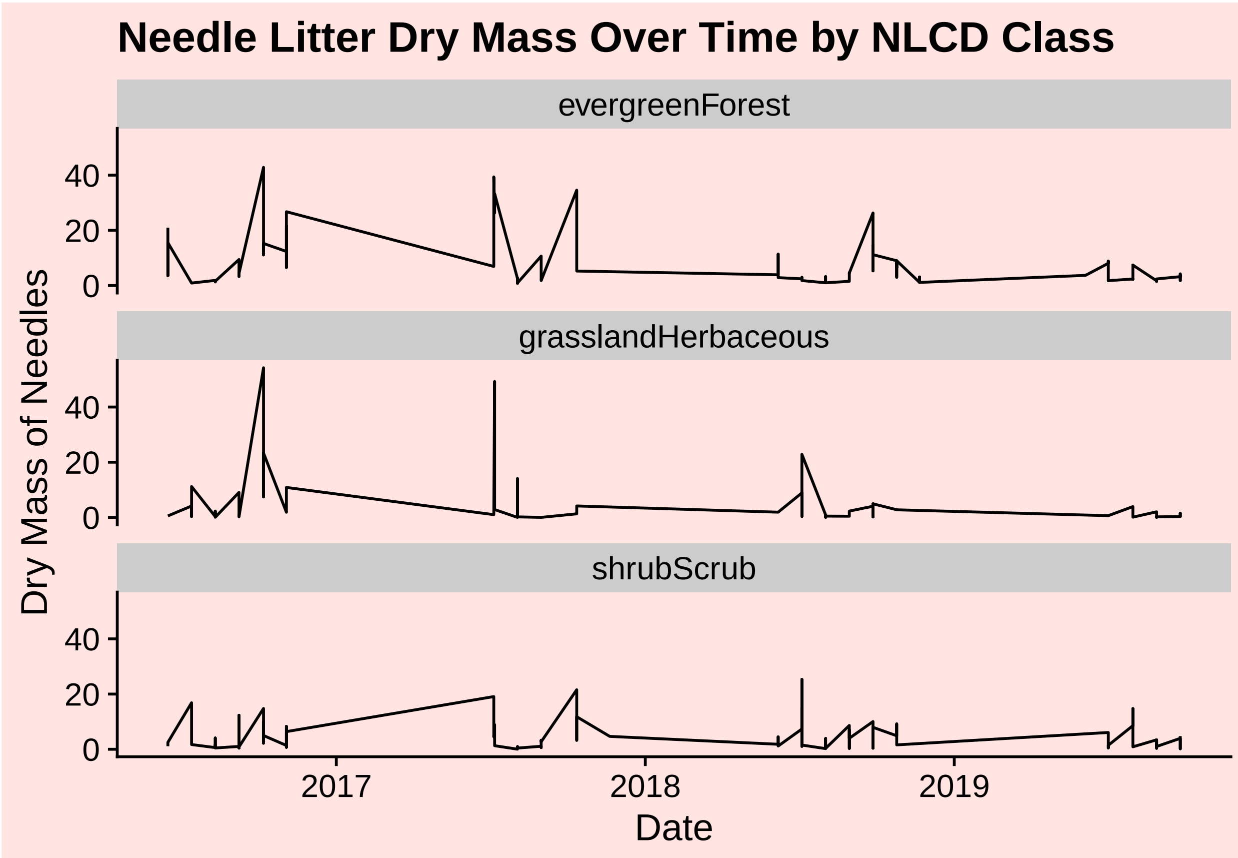
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```r
#6

Needles_subset <- NEON_NIWO %>%
  filter(functionalGroup == "Needles")
```

```r
Litter_Needles <- ggplot(Needles_subset, aes(x = collectDate, y = dryMass,
                                             color = nlcdClass)) +
  geom_line() +
  labs(
    x = "Date",
    y = "Dry Mass of Needles",
    color = "NLCD Class",
    title = "Needle Litter Dry Mass Over Time by NLCD Class"
  )
print(Litter_Needles)
```




```r
#7

Litter_Needles_Faceted <- ggplot(Needles_subset, aes(x = collectDate,
                                                     y = dryMass)) +
  geom_line() +
  labs(
    x = "Date",
    y = "Dry Mass of Needles",
    color = "NLCD Class",
    title = "Needle Litter Dry Mass Over Time by NLCD Class"
  ) +
  facet_wrap(vars(nlcdClass), nrow = 3)
print(Litter_Needles_Faceted)
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: I think the faceted plot is a better visualization because you can more clearly see the pattern of needle dry mass over the years in the three different NLCD Classes. The non-faceted plot was more difficult to analyze because the overlapping lines made it hard to read.